Literature Status: Talagrand's Problem 1 for Boolean Convolution

fa_banach_001 literature-status packet – 19 July 2026

Status. `literature_already_answered` for Problem 1 only. The sharp quantitative Problem 2 remains open in the supporting source.

Original source and question

Gozlan, Li, Madiman, Roberto, and Samson [1] state Talagrand's two Boolean heat-semigroup conjectures in the Introduction on PDF page 2. With μ uniform on $\{-1,1\}^n$, P_τ the heat semigroup, and $f \geq 0$ normalized by $\|f\|_1 = 1$, Problem 1 asks whether

$$\lim_{\eta\to\infty} \eta \sup_n \sup_f \mu\{P_\tau f \geq \eta\} = 0 \qquad (\tau > 0).$$

Problem 2 asks for the stronger sharp estimate

$$\eta \sup_n \sup_f \mu\{P_\tau f \geq \eta\} \leq \frac{c_\tau}{\sqrt{\log \eta}}.$$

The source says on the same page that both conjectures were open.

Supporting answer and identification

Chen's Theorem 1 [2], on PDF page 2, proves that for $\tau > 0$, $n \geq 1$, $\eta > e^3$, and nonnegative f,

$$\mu\{P_\tau f > \eta\|f\|_1\} \leq \frac{c}{(1-e^{-\tau})^2} \frac{(\log\log\eta)^{3/2}}{\eta\sqrt{\log\eta}}.$$

Multiplication by η and passage to the limit immediately gives the limit in Problem 1. More importantly for provenance, the paragraph directly after Theorem 1 explicitly names Talagrand's Problem 1 and states that the theorem resolves it. Thus the supporting author knowingly answers the same problem; the relation is not merely inferred in this packet.

Scope limitation

The factor $(\log\log\eta)^{3/2}$ still diverges, so Theorem 1 does not give the sharp Problem 2 estimate. The supporting paper itself presents its result as resolving the conjecture only up to this dimension-free loss. Accordingly, this packet records a literature answer to Problem 1 only and makes no claim that Problem 2 is solved.

References

[1] N. Gozlan, X.-M. Li, M. Madiman, C. Roberto, and P.-M. Samson, *Log-Hessian and Deviation Bounds for Markov Semi-Groups, and Regularization Effect in L1*, arXiv:1907.10896v2, 2022.

[2] Y. Chen, *Talagrand's convolution conjecture up to log log via perturbed reverse heat*, arXiv:2511.19374, Theorem 1, 2025 (current arXiv version consulted 19 July 2026).